Received: January 05, 2023 | Accepted: January 30, 2023 | Published: February 06, 2023

ORIGINAL RESEARCH

Peercite
Journal of Agriculture and Food Research

THE IMPACT OF FOOD ON THE HUMAN BODY: EXPLORING FUTURE POSSIBILITIES

Charis M. Galanakis

Department of Food Waste Recovery Group, ISEKI Food Association, Vienna, Austria

Citation : Galanakis CM (2023) The impact of food on the human body: exploring future possibilities. Peercite J Agric Food Res. 1(1):6501-6507.

***Corresponding Author : Charis M. Galanakis**

Email :cgalanakis@chemlab.gr

DOI: http://doi.org/10.61641/pjafr.2023.1.1.6501-6507

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https:// peercite.org).

Abstarct:

The world's food system faces major challenges posed by population growth, climate change, geopolitical conflicts, crises, and changing consumer preferences. To address these challenges, it is important to optimize food production, introduce sustainable practices, and develop technological advances while ensuring safety and social acceptance of innovations. This essay covers sustainable food production, food security, climate-resilient and digitalized food supply chains, alternative protein sources, food processing and food technology, the impact of biotechnology, cultural diversity, culinary trends, etc., considers the complex aspects of the future of food. Consumer health and personalized nutrition, the bio economy of circular food production. This article provides a holistic perspective on the evolving food industry, characterized by innovation, adaptability, and a common commitment to the resilience of the global food system. Achieving sustainable, nutritious, and environmentally friendly food production in the future will depend on the food supply chain, including innovative agricultural practices, advances in food processing technologies and Industry 4.0 applications, and approaches that change the way we consume food. Comprehensive changes are needed in various aspects of Redefining food.

Keywords : Food sustainability, Food security, Protein,Bioeconomy, Biotechnology

Introduction :

The global food landscape is changing due to complex developments such as population growth, climate change, a period of condensed crises (including wars, geopolitical conflicts, and pandemic-related impacts), and evolving consumer preferences. Caused by various factors and facing great difficulties.

The world's population is estimated to reach 9.7 billion people by 2050, putting pressure on resource management and making the production of safe, nutritious and sustainable food increasingly urgent. The rapid increase in global food demand, accompanied by urbanization, increased environmental impacts, and agricultural expansion, requires sustainability and resilience of food systems [1].

In this era of challenges and opportunities, debate about the future of nutrition is growing in both the scientific community and society.

Therefore, increasing yields on unprofitable land, increasing crop efficiency, promoting dietary changes, minimizing food waste and optimizing production while reducing environmental impact is important.

Similarly, the food sector requires a transformation consistent with the transition from fossil fuels to biobased products, a climate-neutral economy, digitalization, Industry 4.0 technologies in food logistics and the upgrading of biological resources [2].

Priorities and government policies include agro ecological practices (e.g. rooftop farming, vertical farming, precision farming enabled by IoT applications) and "blue bioeconomy" (e.g. through aquaculture and multi-trophic systems), food safety and digital traceability should be included, equitable food distribution, livestock farming integrated into arable livestock systems and adherence to One Health principles and food safety and digital traceability. Shortening the supply chain through approaches that change the way we consume. Substitutes and experimental meat) will play a central role. V-rich foods and customized food selection based on personalized diets and the role of nutrigenomics and the role of cultural diversity in shaping culinary innovation are important to guide this change.However, other promising innovations, such as Clustered Regularly Interspaced Short Palindromicrepeats and CRISPR-associated Protein 9 (CRISPR-Cas9) gene editing crop improvement, may raise concerns about safety and public perception.There is a gender therefore, consumer preferences should not be underestimated, but rather used as a guide for future food development.

This review article considers multiple aspects that constitute the future of nutrition.This review covers food safety, security and resilience strategies, starting with key aspects such as sustainable food production, alternative protein sources, culinary innovations, the impact of biotechnology, health-related trends, and digitalization of the food chain and their role in circulars. Bioeconomy sustainable food systems. The aim is to provide food researchers with a holistic overview of the dynamic developments the food industry is grappling with by combining insights from these different fields and factors.The latter should be defined by innovation, adaptability and a common commitment to resilient and nutritious global food systems (figure 1).

Figure 1: Illustration of the dimensions that constitute the future of food.

Sustainable food production:

To adequately feed the world's projected population of 10 billion by 2050, global food production will need to increase by at least 70%. But the old practice of simply increasing food production without making the food system more efficient is becoming a thing of the past. Increasing demand for processed foods and depletion of resources have led the industry to place greater emphasis on sustainability [3]. Achieving sustainability is a complex endeavor that requires changes in food consumption patterns, including reducing the link between income and consumption of livestock products, updating existing agricultural models, and making them more resilient to climate change. It includes investments in agricultural systems and active policies for agricultural and agricultural structural development. Food trade regulations.

It also requires reviewing agricultural practices, reducing food loss and waste, maximizing the conversion of raw materials into consumer products, and integrating and coordinating activities across all stages from farm to fork. While certain practices such as reduced tillage, organic farming, and drip irrigation are already well integrated, bio fertilizers, crop rotation, intercropping, and agroforestry are less integrated.

These practices have many benefits and must be adapted to achieve sustainable food systems. For example, innovative approaches to diversifying and intensifying crop rotations within the framework of no-till farming can simultaneously increase crop yields and agricultural profitability while reducing environmental impacts. It is known that This is more pronounced in tropical and subtropical conditions, especially in crops such as soybean, corn, and wheat, where diverse crop rotations outperform double cropping in terms of profitability and gross margins.

Compared to continuous monoculture, diverse crop rotations improve pest and disease control, increase soil fertility and reduce negative impacts on production system performance caused by climate change and biological parameters [4].

Organic farming is any other method that may raise soil health, enhancing nutrient availability and crop high-quality and avoiding chemical inputs (e.g., fertilizers, pesticides, growth hormones, and farm animals feed additives). Progressed soil fertility, biodiversity conservation, carbon sequestration, and weather change mitigation may be finished by promoting agroforestry, which includes deliberately fusing woody elements with lower-degree agricultural manufacturing.

This exercise blessings city regions and densely populated areas via securing a consistent deliver of sparkling, nutritious produce and improving meals machine resilience. Additionally, city agriculture (e.g., rooftop cultivation and indoor farming) integrates water and power resources into evolving urban food systems, aligning with the desires of smart and sustainable cities [5].

Aiming to redefine destiny agriculture, enterprise 4.0 applications, which include synthetic Intelligence (AI), net of factors (IoT), and cyber-physical systems, are remodeling agricultural productivity via facilitating monitoring, precision forecasting, and offering complete insights and machine optimization. For example, AI, especially system gaining knowledge of, is crucial to procedure massive datasets for crop tracking. Those improvements have streamlined data garage and processing, replacing previous techniques. Precision agriculture encompasses the usage of numerous technology (e.g., soil sensors, mapping to satellite-primarily based positioning and connected gadgets, far off manage tools, and automated steering) for extra green operations, reduced aid inputs, and enhanced profitability to offer farmers with extra manage over their fields and teams but, actual-time IoT sensor monitoring improves selection-making efficiency, and its adoption have to deal with protection, complexity, privateness, and reliability issues with ancillary technologies like IoT, unmanned aerial vehicles, field robots, and herbicides, AI revolutionizes weed manage for holistic, facts-pushed methods, reducing herbicide utilization and fostering sustainability [6]. On the other hand, digital twins (DTs) provide superb ability, allowing specialists and farmers to simulate eventualities, test strategies, and expect consequences with precision. This innovation revolutionizes crop control, optimizing resource usage, lowering environmental effect, and enhancing crop yields.

Digitalization of the meals supply Chain:

The conventional meals supply chains face more than one problems, inclusive of food safety, fruit and vegetable losses, disruptions due to worldwide activities, inadequate labeling, and production delays. Food safety and pleasant guarantee challenges encompass detecting alterations in dangerous materials and ensuring authenticity accuracy in cultivation, variety, and geographical starting place. The future of food labeling is predicted to encompass more exact and transparent records approximately ingredients, sourcing, and sustainability, catering to consumers' preference for a greater cognizance of the goods they buy as an instance, digital and QR-code-primarily based labeling systems may also emerge as extra commonplace, providing clients on the spot get right of entry to complete product records via their smartphones [7].

Likewise, challenges like facts change, fabric go with the flow manipulate, economic barriers, and food waste spotlight the need for an in addition digitalization of the supply chain from farm to fork the use of tools including records technologies (Bluetooth, IoT and allotted Ledger era (DLT)), sensors (wi-fi Sensor network (WSN), Temperature and Thermal Imaging (TTI), Barcode, and Radio-Frequency identification (RFID)), region-based totally tech (far flung Sensing (RS), international Positioning machine (GPS), and actual-Time region system (RTLS)), and web apps for the actual-time monitoring and monitoring of food products.

DLTs beautify transparency and agree with among deliver chain members whilst retaining security and privateness, whereas DTs improve the precision of digital representations, allowing system simulation and optimization.

IoT solutions also lessen waste, enhance performance, and enhance visibility in logistics approaches.

Food deliver chains rely closely on collecting and reading statistics for product tracking and stakeholder demands to decorate efficiency, lessen costs, benefit producers, decrease environmental influences, and aid brisker and healthier food transportation.

In the end, future food best is predicted to be signaled via the above technology, imparting clients with actual-time access to records approximately the foundation, manufacturing methods, and protection of meals merchandise but, different challenges should be addressed, inclusive of sustainability issues, authorities' rules, and integrating privateer's renovation with meals traceability [8].

Block chain generation holds sizeable potential to enhance transparency and patron accept as true with within the food supply chain. Its decentralized nature ensures information integrity, making it proof against manipulation.

This era assigns unique digital identifiers to meals products, permitting traceability and preventing fraud, even identifying foodborne illnesses. The blessings consist of statistics sharing inside the industry, immutability, decreased fraud risks, shorter transaction times, and lower charges.

The sizeable traceability is facilitated via a commonplace technological platform, allowing customers to get right of entry to product info via smartphones and providing vital facts on origins, batches, processing, and expiry dates [9].

Nevertheless, implementing block chain in supply chains may be complicated (e.g., lacking standardized procedures), traumatic stakeholder collaboration. Extra research and widespread evaluation fashions are required for seamless integration with the IoT, addressing issues such as meals adulteration and authentication. Moreover, while incorporated with rising technologies, like AI, big records analytics, RFIDs, NFC, IoTs, cloud computing, and analytical methods for the detection of foodborne pathogens, block chain can in addition digitalize food supply chains, enhancing control, automation, efficiency, sustainability, and verifiability. This integration holds transformative promise for the food zone, mainly stores that urge suppliers to embody block chain to decorate supply chain protection and limit food risks. Retail formats which are anticipated to make bigger include e-commerce, pushed by using convenience and personalization, as well as sustainable and moral retail emphasizing eco-friendly practices. Era-pushed shops and direct-to-patron brands additionally constitute growth areas in the retail industry [10].

Alternatively, the usage of AI standards, machines, and deep gaining knowledge of methods has turn out to be an increasing number of universal in studies aimed at analyzing full-size food databases to explore factors like flavors, meals compositions, and chemical compounds. additionally, sensory analysis and flavor checking out carried out via AI systems may additionally play a position in assessing food fine and tailoring tips to individual choices. Indeed, a growth in personalized advertising communications driven by means of facts analytics and consumer alternatives is predicted in food advertising.

On line searching for groceries is likewise anticipated to develop on this course, focusing on improving person reports and incorporating AI-pushed hints to enhance the shopping procedure. Therefore, many meals businesses utilize AI and massive data to enhance product nice, cater to consumer demands, and power the industry in the direction of a cleverer and sustainable future. As an instance, large statistics can be used to investigate consumer perceptions after collecting them from social media platforms to aid fitness food producers in improving their products to align with patron alternatives [1].

Food security and Climate-Resilient food deliver Chain:

Meals safety includes that everybody constantly has bodily, social, and economic get right of entry to an enough deliver of safe and nutritious meals, assembly their nutritional wishes for a healthful life. Meals lack of confidence impacts disproportionately low-income communities and people with health problems and stages from household-degree issues approximately having access to food to excessive infant hunger because of insufficient supply. Meals supply chains (as important infrastructure) and food protection are vulnerable to good sized disruptions, pandemics, and conflict conflicts [1].

Those crises screen the need for studies on optimizing designs and guidelines, encouraging operations, and making use of massive information analytics. More especially, anticipated developments in meals guidelines encompass a stronger awareness on sustainability and environmental effect, with rules promoting eco-friendly practices and reduced meals waste. Health conscious rules may also encourage the reformulation of merchandise to lessen salt, sugar, and dangerous fats, promoting healthier diets even as improving meals protection measures and ensuring transparent labeling to empower purchasers in making knowledgeable alternatives. In addition, supply chain resilience in the agri-meals region can be stepped forward by using customer centric selection-making, proximity-primarily based distribution, and cooperative models, offering an aggressive gain throughout call for fluctuations, as validated at some stage in the COVID19 pandemic [6].

On the other hand, mitigating worldwide warming to restriction it to at least 1.5°C whilst coping with the growing international populace and addressing the increasing demands on land for meals security and carbon sequestration is a complex trouble. Weather alternate and variability pose full-size challenges for smallholder farmers in developing countries (e.g., in African and Asian regions), where many rely upon farming. Extreme and frequent extremes, like droughts, heavy rainfall, temperature shifts, and sturdy winds, have decreased crop yields of key plants (e.g., maize, rice, and potatoes), negatively affecting meals elements, earnings, and food safety for these farmers [3].

Conventional intensive agricultural structures often rely upon heavy resource inputs, monocultures, and heavy chemical use, which pose dangers to soil biodiversity and ecosystems. Unsustainable Intensification methods have caused soil degradation, consisting of acidification, erosion, salinization, and contamination. The sustainable intensification of production on current cultivated land has been proposed as a potential solution to stability those needs through improving agricultural productivity whilst mitigating damaging environmental impacts.

Even though the definition of sustainable intensification has advanced, it basically includes attaining tremendous yield will increase in underutilized or degraded agricultural regions while revitalizing herbal resources. This technique strives to preserve soil biodiversity, reduce nutrient losses, and beautify efficiency.

New strategies for sustainable land control and manufacturing optimization consist of lime to counter soil acidification and improve pH, biochar and zeolites for more advantageous nutrient deliver and water retention, and nitrification inhibitors to reduce N_2O emissions and NO_3 leaching [2].

The destiny of sustainable intensification studies is predicted to prioritize ecological and social blessings, with multiplied involvement from developing nations and ongoing go-nearby collaboration.

However, the extent to which sustainable intensification can boom yields remains unsure. Sustainable intensification practices in South Asia yielded a median increase of 21%, with residue retention and natural fertilizers displaying wonderful advantages. Profitability turned into no longer constantly guaranteed, and yield gains have been modest in comparison to the prevailing yield gaps [1].

Aiming to increase meals safety, environmental preservation, productivity, and manufacturing range in a converting weather, smallholder farmers ought to undertake preparedness strategies to reduce vulnerability and crop losses. Those techniques ought to emphasize access to agricultural inputs, market statistics, crop diversification, irrigation, and weather-smart tech even as accounting for network perspectives. Addressing climate change influences calls for intensified cooperation among research institutions, policymakers, crop managers, and farming communities. Sustainable practices, like weather-smart soil control and drought-resistant crop breeding, becomes essential in the following years.

Farmer involvement, government regulations, and subsidies are essential for environmentally friendly crop production, limiting annual area motion and mitigating deforestation. Effective coverage implementation would require an extra holistic technique considering productiveness and environmental sustainability in diverse agricultural systems. Additionally, multispectral collaborations and multilevel interventions are deemed necessary to mobilize assets for addressing the mixed impacts of climate change, meals lack of confidence, and the pandemic, especially in decrease- and center-earnings nations [9].

Furthermore, incorporating carbon sequestration and biodiversity into sustainable intensification dreams can enhance its relevance and mitigate the point of interest on productiveness and yields. though under risk, biodiversity plays a pivotal position in mitigating climate alternate by means of serving as a natural defense mechanism against severe climatic activities and freeing greenhouse gas (GHG) emissions. Soil biodiversity is crucial to surroundings features, consisting of soil aggregation, carbon sequestration, natural be counted decomposition, and nutrient cycling. As a result, its protection is turning into very important for the meals sector, and unique actions are required on this direction. For example, Crop Wild loved ones are wild plant species intently associated with cultivated plants and possess precious genetic traits that decorate crop productiveness and resilience in changing environmental conditions [2].

Alternative Protein sources:

Dietary proteins are pivotal in human nutrition and addressing food protection demanding situations. Sooner or later, the global protein factor market is continuously growing, mirroring the escalating demand and the depletion of important sources pushed by populace increase.

Although animal-based totally proteins possess a big nutritional satisfactory, extended meat consumption is related to moral challenges and significant environmental concerns due to expanded GHG emissions contributing to climate trade. Alternatively, diets that encompass better amounts of opportunity proteins and lower meat intake are more and more being diagnosed for their tremendous effects on the environment and human fitness, e.g., the consumption of plant proteins is related to a reduced continual ailment danger [10].

In reaction to these demanding situations, researchers actively pursue sustainable protein sources that yield nutritional and economically feasible merchandise. This isn't always always an easy challenge since animal proteins are regarded as entire because of their content material of important amino acids, a characteristic not shared by using all plant proteins, which are often deemed incomplete because of their deficiency in positive amino acids (e.g., lysine, threonine, and sulfur-containing amino acids like cysteine and methionine) required for proper human boom. Positive plant vegetation, like soybeans, peas, quinoa, and amaranth, comprise all the essential amino acids.

Other plant-based totally proteins, like oats, lupin, or wheat, successfully meet amino acid desires while combined with animal proteins. Protein blends can mimic animal-based protein traits, and in this experience, connecting plant (e.g., wheat, oat, lupin, and wheat) and animal proteins can assure adequate essential amino acid intake. Similarly research to enhance vitamins, simulate flavors, and improve protein functionality is essential to appeal to substitutes for conventional meat merchandise [3].

Cultured meat presents any other promising alternative protein supply. Research is now directed in the direction of the sustainability of lab-grown meat production and purchaser recognition, because the success of these products is predicated on replicating meat's flavor. Purchaser preferences regarding exclusive meat types and scalability in developing international locations and regions characterised by using low meat consumption levels are different tremendous challenges that should be addressed. Beyond traditional meals markets, lab-grown meat should discover a market in pet food and numerous non-meals products, decreasing poaching [6].

Though, cultured meat confronts several technical demanding situations, such as replicating meat diversity across animal species, breeds, and cuts. Protection blessings from cultured muscle cells exist but issues about dysregulation and nutritional composition persist. There may be also an ongoing debate about the environmental benefits of GHG emissions, extensive opposition from plant-primarily based alternatives, and uncertainty surrounding the religious acceptability of cultured meat, whether it aligns with Kosher or Halal nutritional requirements. Furthermore, regardless of ongoing studies, the future competitiveness of cultured meat in opposition to conventional meat and substitutes stays unsure as policymakers face strain to alter cultured meat production. Policymakers have to use those insights for tailored advertising and marketing strategies considering societal and monetary factors. Powerful verbal exchange, scale-up manufacturing, actual-time sensing for nutrient recycling, and regulatory tips are essential components to be addressed by policymakers [8].

Fit for human consumption bugs gift another sustainable opportunity source of protein and other vitamins, like fats and minerals, imparting the potential for improving several nutrition objects, along with snacks, pasta, protein bars, and bread. Almost 2000 edible insect species (grasshoppers, bees, caterpillars, crickets, locusts, wasps, beetles, and ants) exist globally, normally consumed inside the tropical and subtropical regions of Africa, Asia, and the Pacific areas, however encounter confined popularity in Western international locations.

Insect proteins are nutritionally precious, with practical residences like gelling, foamability, and emulsification. Additional benefits consist of the year-round availability and low environmental effect as compared to meat proteins [7].

The studies into extraction techniques (e.g., wet fractionation and non-thermal techniques, like ultrasound and microwave remedies) and bioactive peptides from insects is ongoing, and procedures consisting of using microbial fermentation, simulating gastrointestinal digestion, and applying business enzymes are taken into consideration for the extraction of antidiabetic, antioxidant, and anti-hypertensive peptides. Nevertheless, the wider development of the insect marketplace faces challenges that want to be surpassed, e.g., defatting methods for boosting insect protein capability, allergen city, protection troubles, patron popularity, willingness to pay, and clear policies. Collaborating with healthcare professionals is usually recommended, and in addition measures can be important to make certain the manufacturing of high-quality prebiotics from insects [3,7].

Algae's excessive protein content and low environmental effect lead them to any other promising protein source to produce supplements and food components in sweets, bread, noodles, liquids, and beer, as well as meat analogs with a fibrous texture (the use of excessive moisture extrusion) and nutraceutical claims related to bioactive peptides and antioxidant capacity.

Numerous extraction strategies can efficaciously isolate algal proteins for nutritious and coffee-fee meals production, although there's still uncertainty approximately their bioavailability, necessitating further in vivo research research. Subsequently, in addition studies is wanted to improve protein extraction and purification methods and non-thermal technologies can be used on this path for instance, ultrasound-assisted extraction has been used to decorate the yield of antioxidants from diverse assets, together with microalgae, imparting shorter extraction instances, reduced solvent consumption, and lower temperatures [6].

Safe to eat wild mushrooms and single-cell proteins produced by fungi also are emerging as opportunity protein resources due to their efficiency in converting waste substances into protein-wealthy biomass.

Though, nucleic acid content material, allergenic capacity, other safety worries, standardized manufacturing tactics, and regulatory frameworks need to be addressed. Mycoprotein, derived from filamentous fungi, is another value-powerful and nutritionally useful opportunity protein that serves as a sustainable meat alternative, offering vital nutrients performing as a prebiotic, antioxidant, and regulator of ldl cholesterol, blood glucose, and muscle protein improvement. Sooner or later, microbial proteins produced the usage of air show low land and water requirements decreased GHG emissions, and the capacity to fight international malnutrition, aligning with worldwide sustainability goals [4]. As an instance, solar foods (a Finnish startup) produces a carbon-neutral microbial protein the use of CO_2 from air, water, and minerals. The so-called "Solein" product offers versatile technofunctional residences in food applications, matching the homes of plant-primarily based protein isolates. Its regulatory reputation varies by area, with the ecu's Novel food status making sure protection, at the same time as other regions have their regulatory approaches, like GRAS inside the USA. Similarly, studies need to comprehensively evaluate alternative proteins, considering distinct consumer segments to improve popularity, encouraged through attitudes, familiarity, taste, disgust, food neophobia, and social norms [8].

Conflict of interest :

Author declares no conflict of interest.

References :

1. Galanakis CM (2020) The Food Systems in the Era of the Coronavirus (COVID-19) Pandemic Crisis. Foods. 9:523.

2. Galanakis CM, Rizou M, Aldawoud et al. (2021) Innovations and technology disruptions in the food sector within the COVID-19 pandemic and post-lockdown era. Trends Food Sci Technol. 110:193-200.

3. Foley JA, Ramankutty N, Brauman KA et al. Solutions for a cultivated planet. Nature. 478:337-342.

4. Kamilaris A, Fonts A, Prenafeta-Boldu FX (2019) The rise of blockchain technology in agriculture and food supply chains. Trends Food Sci Technol. 91:640-652.

5. Galanakis CM (2022) Sustainable applications for the valorization of cereal processing by-products. Foods. 11:241.

6. Galanakis CM, Brunori G, Chiaramonti D et al. (2022) Bioeconomy and green recovery in a post-COVID-19 era. Sci Total Environ. 808:152180.

7. Rizou M, Galanakis IM, Aldawoud TMS et al. (2020) Safety of Foods, Surfaces, Food supply chain and environment within the COVID-19 pandemic. Trends Food Sci Technol. 102:293-299.

8. Godfray HCJ, Beddington JR, Crute IR et al. (2010) Food security: The challenge of feeding 9 billion people. Science. 327:812-818.

9. Van Zanten HH, Van Ittersum MK, De Boer IJ et al. (2019) The role of farm animals in a circular food system. Glob Food Secur. 21:18-22.

10. Galanakis CM, Aldawoud TMS, Rizou M et al. (2020) Food ingredients and active compounds against the coronavirus disease (COVID-19) Pandemic: A comprehensive review. Foods. 9:1701.